Low-Level Windshear Detection from Passively Received Mode S EHS at Helsinki-Vantaa

Development phases, algorithm changes, their reasons, and the database checks of the EFHK study

Otso Laakso (OH2GAX)
Tuusula, Finland · github.com/oh2gax/mode_s_wind

Abstract. MODE-S Wind is a single passive 1 090 MHz receiver near Helsinki-Vantaa airport (EFHK) that decodes the Mode S Enhanced Surveillance (EHS) replies of aircraft to the ground interrogators. On final approach and on the climb-out every equipped aircraft reports its true airspeed (TAS) and ground speed (GS) about every two seconds; their difference is the headwind the aircraft is flying into. This log follows an attempt to turn that stream into an objective, per-flight report of low-level windshear — something that, to our knowledge, has not been published for passively received Mode S data. The study started with six simple browser algorithms (May – October 2026). A replay of 3 084 landings showed that they mostly reacted to the normal change of the wind with height, and on 24 October 2026 they were abandoned in favour of the headwind-ramp method of the Hong Kong Observatory (HKO), scaled by the length of the ramp, together with a comparison of each flight's 200-ft band winds with the flights of the previous 45 minutes. The log then records each later change — a relative rule for weak ramp advisories (26 October), a label that tells shear seen by several aircraft from single-aircraft turbulence (27 October), and a flight-phase correction of the band comparison (27 October) — with the evidence that led to it and its replay on the stored data.

Research only. All results come from a passive, uncertified receiver and describe air that an aircraft has *already* flown through. They are not an operational windshear warning and must not be used for flight operations or air traffic control.

Contents

About this document 2

1 Setting and data 3

2 Phase 0 — six browser algorithms (May – 24 October 2026) and why they were abandoned 3

3 Phase 1 — detector v2 (24 October 2026) 5

4 Phase 2 — relative rule for weak ramp advisories, v4 (26 October 2026) 7

5 Phase 3 — confirmed or isolated (27 October 2026) 7

6 Phase 4 — band comparison by flight phase, v5 (27 October 2026) 8

7 Phase 5 — higher band limits for turboprop departures, v6 (27 October 2026, evening) 9

8 New tools for the study (27 October 2026, late evening) 11

9 The first night of the departure trial: a nocturnal low-level jet (28 October 2026) 11

10 Comparison with the Hong Kong lidar method (28 October 2026, evening) 13

11 Phase 6 — a windshear alert needs a 14-kt change, v7 (28 October 2026, late evening) 15

12 Turbulence (EDR) from the replies (29 October 2026) 15

13 The boundary layer: low-level jet and surface inversion (29 October 2026) 17

14 Validation data: METAR archive and pilot reports (29 October 2026) 19

15 Phase 7 — the ADS-B ground speed and the BDS 6,0 airspeed, v8 in shadow (30 October 2026) 19

16 Results so far 20

17 Database checks 21

18 Current method and parameters (version v7) 22

19 Limits, open questions and next steps 22

About this document

This is a *living* study log. It is written for aviation meteorologists and atmospheric scientists who want to follow how the method develops, and it is updated whenever the detection changes or a new check of the database has been made. The rules for keeping it are:

- Each change of the method gets its own phase section: what was changed, *why*, the evidence behind it, and the replay of the stored flights with the new version. Earlier sections are not rewritten; a later correction is marked as such.
- Every check of the database is added to the dated log in Section 17, also when nothing was changed.
- The current parameters are always in Section 18, the open questions in Section 19.

Times are UTC. Heights are feet above mean sea level (MSL) unless said otherwise; the EFHK thresholds are at about 180 ft. "kt per 1 km" means the 1-km equivalent of a headwind ramp (Section 3.2). The user-facing description of the present detector is the *Windshear User Guide* [26]; the retired algorithms are documented in [25].

Table 1. Version history of this document.

Version	Date	Content
1.0	27 Oct 2026	First edition: Phases 0–4 (May – 27 October 2026), the database checks of 24–27 October 2026, parameters and open questions.
1.1	27 Oct 2026, evening	Phase 5: higher band limits for turboprop departures (v6); the evening data check of 27 October; results and parameters updated to v6.
1.2	27 Oct 2026, late evening	New tools (Section 8): the space–time diagram of the band changes, the raw climb-out series and the trial ramps along the departures; next steps updated.
1.3	28 Oct 2026	The first night of the departure trial (Section 9): a nocturnal low-level jet — the trial ramps flag the normal climb into it, and two departure advisories during its weakening; the data check of 28 October; two questions added to the next steps.
1.4	28 Oct 2026, evening	Comparison with the Hong Kong lidar method (Section 10): what the HKO papers say, a replay of GLYGA's 14-kt alert rule over all stored landings, the severity–length (Woodfield) diagram and the alert duration; the data check of 28 October evening; a proposal for v7.
1.5	28 Oct 2026, late evening	Phase 6 (Section 11): the 14-kt rule implemented as v7; results and parameters updated to v7.
1.6	29 Oct 2026, evening	Three new research products beside the detection: the turbulence (EDR) of every flight (Section 12), the boundary layer — low-level jet and surface inversion — hour by hour (Section 13), and the METAR archive and pilot reports as validation data (Section 14); how each relates to the windshear flags; the data check of 29 October; next steps updated.
1.7	30 Oct 2026	Phase 7 (Section 15): the ADS-B ground speed and the BDS 6,0 airspeed in the headwind — detection v8, running in shadow beside v7 for a week; the check of 30 October; parameters and next steps updated.

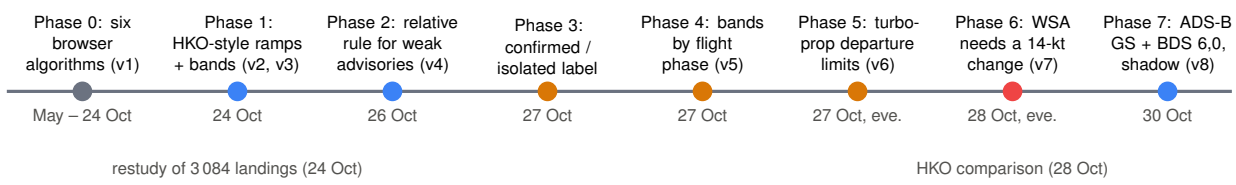


Figure 1. The phases of the study (not to scale). The stored report version (v) is given in brackets.

1 Setting and data

1.1 Receiver and airport

The receiver (a Jetvision Radarcape with a ground-plane antenna) is in Tuusula, north of EFHK, and feeds a Raspberry Pi 4 that decodes, stores and analyses the data. EFHK has three runways: 04L/22R, 04R/22L and 15/33. The Finnish surveillance network — Mode S radars and the wide area multilateration (WAM) system [14, 15] — interrogates the aircraft; MODE-S Wind only listens to the replies.

1.2 The Mode S EHS registers used

The Comm-B replies (DF 20/21) carry the requested register, without its identifier [3]. Two registers matter here:

- **BDS 5,0, track and turn report**: roll angle, true track, ground speed, track angle rate and **true airspeed**. GS and TAS have 2-kt resolution.
- **BDS 6,0, heading and speed report**: magnetic heading, indicated airspeed, Mach number and two vertical rates.

The wind vector of an aircraft follows from a BDS 5,0 / 6,0 pair (ground vector minus air vector, with a heading calibration learnt per airframe) [12]. For windshear the more useful quantity is simpler: **TAS − GS from one BDS 5,0 reply**. It needs no heading, both speeds come from the same moment, and it equals the along-track headwind plus a small, slowly varying crab term $\mathrm{TAS}(1-\cos\beta)$ (about 2 kt at a 10° crab angle β), which cancels in changes along one flight.

1.3 What is stored

- Every observation (position, altitude, decoded registers) since 24 September 2026.
- **Raw approach series** since 9 October 2026: for every landing the BDS 5,0 and 6,0 replies and the 3-s position sweeps from about 12 NM to the lowest point seen. Since 24 October also the ADS-B velocity messages (ground speed at about 2 Hz).
- **Band profiles** of every landing and, since 11 October, every departure: the mean wind vector, TAS − GS and the number of wind pairs in 200-ft bands.
- **A windshear report** (`shear_json`, about 0.35 kB) for every landing and departure since 24 October 2026; older flights were back-filled from the stored data. Every change of the method raises the report version, and the back-fill then recomputes all stored flights once, so the counts in this log always refer to one method over the whole period.
- For comparison: the MEPS model of MET Norway [13] (optional; nothing in the detection depends on it), the Jokioinen soundings, and the EFHK METARs.

Table 2. Data quality on final approach, 3 084 landings 9–23 October 2026 [16].

Quantity	Result
BDS 5,0 reply interval	median 1.97 s (p90 3.95 s), about 144 m along the path
Interval between changes of TAS − GS	median 3.96 s (about 290 m, 50 ft on a 3° path); 23 % of the replies repeat the previous register
Per-reply headwind noise (RMS)	median 1.06 kt (p90 1.50); the 2-kt quantisation alone gives 0.82 kt
Lowest reply above the threshold	15: 232 ft · 22L: 337 · 22R: 349 · 04L: 508 · 04R: 733 ft (median by runway; set by the receiver's line of sight)
Largest 1-km headwind loss per approach	median 4 kt, p99 8 kt (smoothed profile, benign weather)
Horizontal F-factor over 1 km	median 0.014, p99 0.03, maximum 0.065 (threshold 0.105 never reached)

A 15-kt change over 0.4–4 km is therefore about ten times the noise of the difference of two 400-m means. The practical floor is set by the natural variability of the wind, not by the data.

2 Phase 0 — six browser algorithms (May – 24 October 2026) and why they were abandoned

2.1 What they were

The first windshear view ran in the browser of each viewer, only while the page was open. It polled the latest heading-calibrated wind of each aircraft in the approach corridor every 3 s, turned it into a headwind on the runway heading and applied one of six algorithms, chosen per browser. Levels were Monitor 10 kt, Warning 15 kt and Alarm 25 kt (loss or gain), with a two-poll confidence gate. Full details are in [25].

Table 3. The six v1 algorithms, their main problem, and the share of landings that would have reached Warning ($\geq$ 15 kt) in 14 benign days (9–23 October 2026; highest METAR wind 17 kt, no windshear reported) [16].

Algorithm	Compared	Main problem	Warning
Energy	GS + alt/100 over 45 s	ground speed instead of airspeed; 1 kt equals about 12 ft at approach speed, not 100 ft; the planned descent and deceleration look like an energy loss	97.9 %
Gradient	largest headwind difference of two points 200–3 000 ft apart	spans up to 17 km of path; the maximum over all pairs grows with noise; no smoothing, no length scaling	13.3 %
Pairwise	two aircraft 200–2 000 ft apart	measures the normal wind profile plus two airframes' own TAS and heading biases	7.6 % of pairs
Rate	current headwind vs older stored points	look-back up to 2 000 ft / 3 NM without length scaling	3.8 %
Kinematic	IAS − GS over 45 s	IAS is not TAS (about 4 kt at 2 000 ft falling to 1 kt at 500 ft), so the descent itself fakes a headwind change	2.3 %
Baseline	headwind vs the mean wind below 2 000 ft of the last 30 min	mixes all heights into one expected value	0.2 %

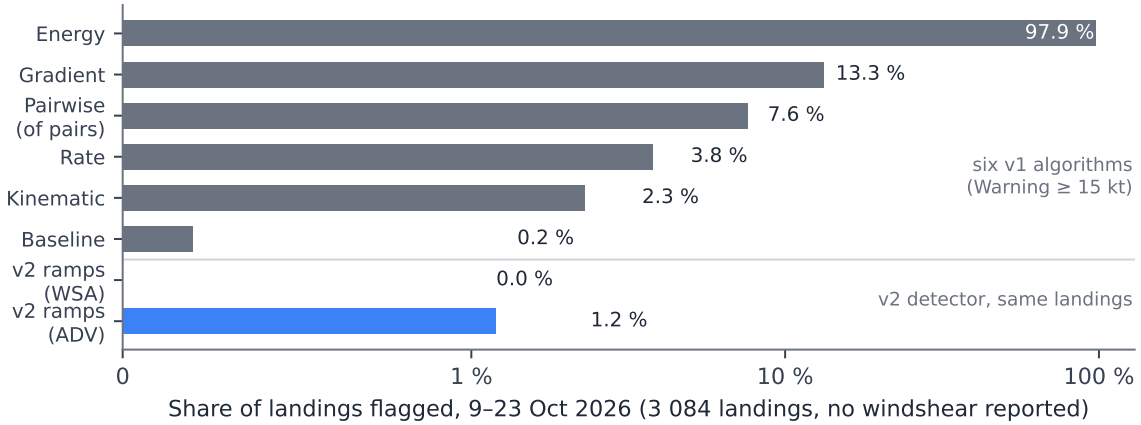


Figure 2. Share of landings flagged in 14 benign days by each v1 algorithm (Warning, grey) and by the v2 ramp detector on the same landings (blue: advisory, orange: windshear alert). Logarithmic axis.

2.2 The common faults

- **No scaling with the distance of the change.** A 15-kt change means very different things over 400 m and over 10 km. Every operational method — the HKO glide-path ramps, the 1-km F-factor, the station spacing of LLWAS — ties the threshold to a length or a time [4, 8, 10].
- **Altitude instead of distance along the path**, which mixes shear along the path with the ordinary increase of the wind with height — the "gentle ramps" that HKO filters out [6].
- **Loss and gain treated alike**, although the headwind loss is the performance-decreasing case, and an Alarm level (25 kt) that matches no standard.
- **One poll's wind per aircraft**, noisier than the TAS − GS of the replay.

Kept from v1: the gating of the corridor and of an established final, the go-around detection and the guard against frozen GNSS positions. The decoded BDS 4,5 hazard flags were also examined: 47 % of them showed a windshear level above 0 in a period with no windshear, so they are other registers misidentified as BDS 4,5 and must not feed any alert.

2.3 The decision: one method after the Hong Kong Observatory

On 24 October 2026 all six algorithms were retired at once (no parallel "legacy" mode). The replacement follows the GLYGA method of the Hong Kong Observatory, which finds headwind *ramps* along the glide path in Doppler-lidar scans and alerts when $\Delta V / L^{1/3}$ exceeds a threshold for ramp lengths L of 400–4 000 m [4, 5, 7]. The reason for choosing it is that the lidar's input — a one-dimensional headwind profile along the approach path — is exactly what each approaching aircraft samples with its own BDS 5,0 replies. The $L^{1/3}$ scaling comes from flight-data studies of how an aircraft responds to a change of headwind, so it carries aircraft-performance meaning that the v1 thresholds lacked.

HKO has also noted that Mode S downlinked winds are, in principle, fast enough for real-time windshear alerting [16], but no evaluation has been published; this study tries to provide one.

3 Phase 1 — detector v2 (24 October 2026)

The detection moved to the server. It runs for every landing and departure whether or not anyone has the page open, and each flight gets a stored report.

3.1 Headwind profile along the path

For each reply TAS − GS is computed. Replies are dropped when the roll exceeds 5°, when the register repeats (same GS and TAS within 6 s), or when they are more than 12 kt from the running median of ±7.5 s. The distance along the path comes from the integrated ground speed. The profile is a 400-m running mean on a 100-m grid.

3.2 Headwind ramps

For every grid point and length $L \in \{400, 600, 800, 1\,000, 1\,500, 2\,000, 3\,000, 4\,000\}$ m the change ΔV over L is scaled to a 1-km equivalent

$$\Delta V_{1\mathrm{km}} = \Delta V \left(\frac{1\,\mathrm{km}}{L} \right)^{1/3}, \tag{1}$$

and the strongest loss and the strongest gain of the flight are kept. A ramp counts only if at least 60 % of it is covered by data and $|\Delta V|$ is at least five times its own noise; ramps of 600 m or less need three replies at each end and must persist over the next 400 m. Only ramps in the low-level zone count: below 1 600 ft above the threshold or within 5 NM of it [1, 2]. The levels, in kt per 1 km:

ADV advisory ≥ 10 **WSA** windshear alert ≥ 15 **MBA** microburst alert ≥ 30 (loss only)

Beside the ramps, the horizontal F-factor $F_h = \dot{W}_x/g$ is computed as a mean over 1 km [8, 9]; the vertical term $-w/V$ cannot be observed from Mode S.

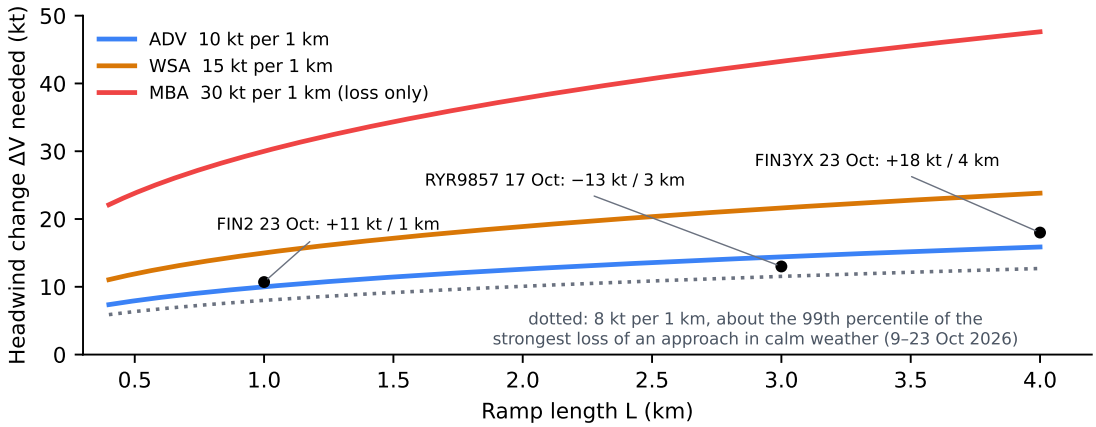


Figure 3. The headwind change needed for each level as a function of ramp length. The dotted line is about the 99th percentile of the strongest loss of an approach in calm weather.

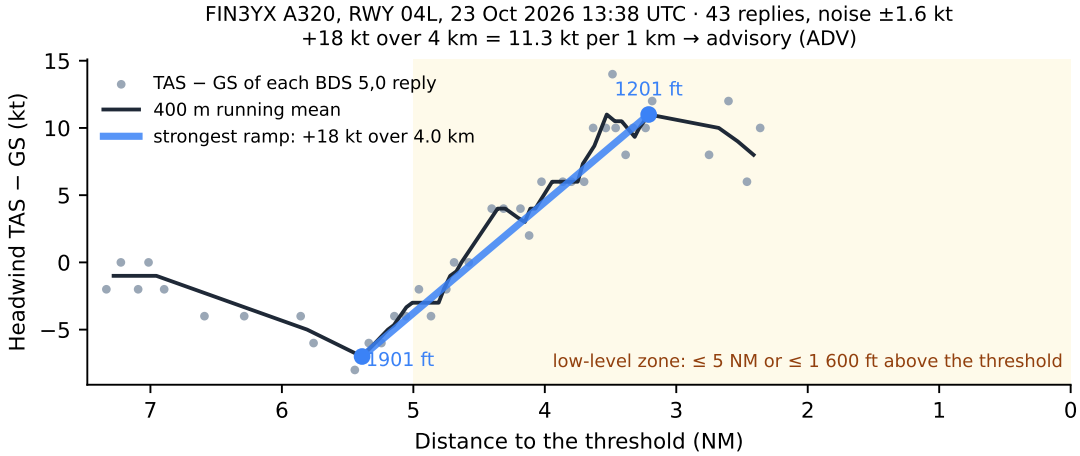


Figure 4. A ramp in practice: the per-reply headwind, the 400-m mean and the strongest ramp of one landing (23 October 2026, a persistent shear layer before a front).

3.3 Band comparison between flights (v2 → v3 the same day)

A ramp shows what one aircraft met. A front or gust front shows as a change *between* flights. For this the wind of each 200-ft band from 400 to 1 800 ft is compared with the median of the flights of the previous 45 minutes (at least 3 references); a change of ≥ 10 kt is ADV and ≥ 15 kt WSA.

The first version (v2, morning of 24 October) compared the band means of TAS − GS with *landings on the same runway heading*. A check the same afternoon [17] showed that this did not work for departures: 55 % of departures had no landing on the same runway direction in the previous 45 minutes (the usual configuration is landing 15, departing 22R), and climb-outs give too few replies within ±15° of the axis.

Why the change. TAS − GS is tied to the flight's own direction, so it can only be compared with flights in the same direction. The **wind vector** of each band (from the BDS 5,0 / 6,0 pairs) can be compared across all runways. In v3 (evening of 24 October) the reference became the median u, v of all flights of the last 45 minutes — landings and departures — and the difference is projected on the flight's own direction:

$$\Delta H_b = -\left(\mathbf{w}_b - \mathrm{med}_{\mathrm{ref}}\,\mathbf{w}_b\right) \cdot \hat{\mathbf{d}}, \tag{2}$$

where $\hat{\mathbf{d}}$ is the unit vector of the flight direction and a negative ΔH_b is a headwind loss. Bands with corrupt values (speed above 80 kt or spread above 15 kt) are skipped. Adding departures raised the share of landings with at least 3 references from 88.9 % to 92.5 % and filled the night gaps when there is about one landing per hour.

3.4 First results

Table 4. v2/v3 replay on the stored flights (benign weather apart from 17 October).

	Flights	ADV	WSA
Landings 9–23 Oct, ramps	3 084	37 (1.2 %)	0
Landings 9–24 Oct (v3), ramps / band	3 310	38 / 0	0 / 4
Departures 11–24 Oct (v3), band	3 084	21 (0.7 %)	9 (0.3 %)

Most ramp advisories were real low-level wind-profile cases, such as the night inversion of 11 October (+17 kt over 3 km between 1 800 and 1 200 ft in successive landings) and the persistent shear layer before the front of 23 October. The band WSAs were the wind change of **17 October 2026** on runway 22L (Figure 5): FIN2EZ, RYR9857 and FIN2AT at 09:04–09:07 had 25–29 kt less headwind at 800–1 200 ft than the landings ten minutes earlier. The departure FIN863 from 22R had already shown −24.7 kt at 1 200 ft at 08:54, *ten minutes before the first landing*; the departures that switched to runway 04 at 09:13–09:38 then saw the wind reversal (+20…+38 kt). The fourth band WSA was on 18 October at 08:41, and a departure on 24 October at 04:42 showed +16 kt at 1 800 ft during the night frontal passage. The noise of the reply stream stayed at a median of 1.24 kt.

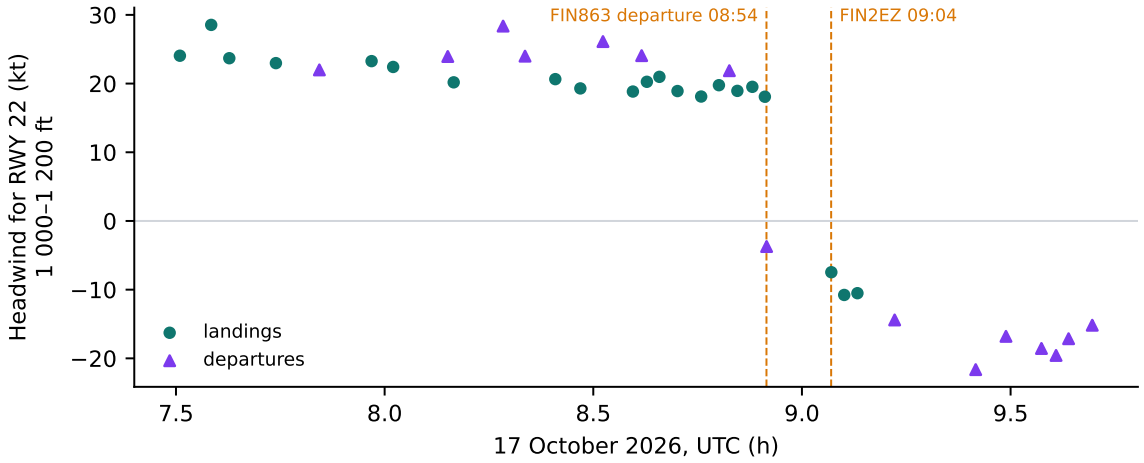


Figure 5. The 17 October 2026 case: band headwinds of the landings on 22L before and after the wind change.

4 Phase 2 — relative rule for weak ramp advisories, v4 (26 October 2026)

4.1 Observation

On the evening of 25 October a stable boundary layer formed under a strong southerly flow: the median headwind of the landings rose from 20 kt at 400 ft to 35 kt at 1 800 ft. Between 20:00 and 22:30 UTC 10 of 39 landings got a ramp advisory. Each ramp was real — the aircraft did lose that headwind — but every aircraft lost the same amount at the same heights, and the band comparison saw nothing (largest change 7.6 kt). The advisory flagged the expected profile, not anything unusual. In the whole period 9–26 October, 59 landings (1.6 %) had a ramp advisory, all at 10–14 kt per 1 km [18].

4.2 Change

A ramp of 10–12 kt per 1 km stays an advisory only if it exceeds the change that the reference flights had over the same heights by at least 10 kt per 1 km. With $H_{\mathrm{ref}}(a)$ the median reference headwind at altitude a (from the band references of Section 3.3) and a_1, a_2 the ends of the ramp:

$$e = H_{\mathrm{ref}}(a_2) - H_{\mathrm{ref}}(a_1), \qquad r = \frac{|\Delta V - e|}{|\Delta V|}\,\Delta V_{1\mathrm{km}} \quad (\text{when } \Delta V - e \text{ has the sign of } \Delta V, \text{ else } r = 0), \tag{3}$$

and the ramp is ADV if $\Delta V_{1\mathrm{km}} \geq 12$ or $r \geq 10$. Without references (at night) the absolute rule applies. **WSA and MBA stay absolute.**

> **Why the change.** The advisory is meant to point at something the other aircraft did *not* have. The ordinary increase of the wind with height in a stable layer is a known, "gentle ramp" situation that HKO also removes from its alerts [6]. The stronger levels are kept absolute because a 15-kt-per-km loss matters to the aircraft whether or not the previous aircraft had it too.

Replay 9–26 October: landing advisories 59 → 26; in the windy days 24–26 October 21 → 13. The band WSAs (4) and the departures were unchanged.

5 Phase 3 — confirmed or isolated (27 October 2026)

5.1 Observation: the turbulent southerly day of 26 October

26 October brought a strong, warm and moist southerly flow: surface wind 12–17 kt (gusts to 28–33 kt at 02–03 UTC), 30–40 kt at 1 000–4 000 ft, and a near-neutral lapse rate in the Jokioinen soundings — mechanical turbulence. The Approach Conditions Index of the landings had its highest median so far (4.2; typically 1.1–2.2) and the per-reply noise rose from about 1.2 to 1.86 kt [19]. Of 200 landings in 24 hours, 12 had a ramp alert (9 ADV, 3 WSA). Looking at each case with its neighbours:

- **Real shear layer:** FIN6KB at 13:08 on 22L. Every 22L arrival lost 7–10 kt of headwind at 1 150–1 350 ft, and four neighbours had a matching ramp.

- **Partly confirmed:** FIN7SM at 14:06 (WSA): an 11-kt ramp over 400 m at about 1 100 ft; FIN7RL had the same dip at 1 000 ft three minutes earlier — a real, short eddy or gust.
- **Isolated:** seven landings (two of them WSA) had a short ramp that no neighbour within ±15 minutes repeated, while their 200-ft band profiles looked like their neighbours'. These are turbulence rather than windshear.

5.2 Change

A ramp alert is labelled **confirmed** when another landing on the same runway direction within ±15 minutes had a ramp of the same kind (loss or gain) of at least 7 kt per 1 km whose middle lies within ±300 ft of this one's; otherwise **isolated**. The label is first computed from the landings before and completed 15 minutes later from the landings after. *The alert level is not changed.*

> **Why the change.** A shear layer, front or gust front is met by several aircraft; a turbulent eddy over rough terrain by one. The label lets the reader — and later statistics — separate the two. The level is kept because the first aircraft into a new shear is always alone at that moment; the 17 October departure FIN863 is the example.

Result on the stored data (9–27 October): of 36 landings with a ramp alert, 25 are confirmed and 11 isolated; 10 of the 11 isolated ones are from the turbulent day of 26 October (Figure 14).

6 Phase 4 — band comparison by flight phase, v5 (27 October 2026)

6.1 Observation: departures read more headwind than landings

Already in the first version a constant offset had been seen: in the same bands, departure TAS − GS was 2.3 kt above that of landings (1 946 replies, 9–23 October), and v2 simply subtracted it. When the band comparison moved to the wind vector (v3) the correction was dropped, but the offset remained: departures read +1.4…+1.7 kt more headwind than the landings within ±20 minutes (4 543 band observations), and in the windy night of 26–27 October +4…+9 kt. Of the 36 departure band flags up to 27 October, 27 were "more headwind" and only 9 "less".

6.2 Tests

- **Acceleration (a lag between the GS and TAS sources):** not the cause. Among the departure observations the offset hardly depends on the acceleration (slope −0.64 kt per kt s^{-1}, $r = -0.10$), and in the raw landing series the *steady* segments have the largest offset (−1.5 kt) while decelerating ones have about 0.
- **Geography (departures and landings fly through different air):** not the cause. Departures to 04 read +1.7 kt and to 22 +1.4 kt.
- **Comparison with the MEPS model** at 600–1 800 ft (model levels, pressure from QNH), headwind along each flight's own direction: landings −1.45 kt, departures +1.38 kt (Table 5, Figure 6). The offset is *symmetric about the model*, and the wind speeds of departures agree with the model exactly.

Table 5. Measured minus MEPS area-mean wind, 600–1 800 ft, 9–27 October 2026.

Flights	Band means	Headwind − MEPS	Wind speed − MEPS
Landings	9 439	−1.45 kt	−0.9 kt
Departures	6 789	+1.38 kt	0.0 kt

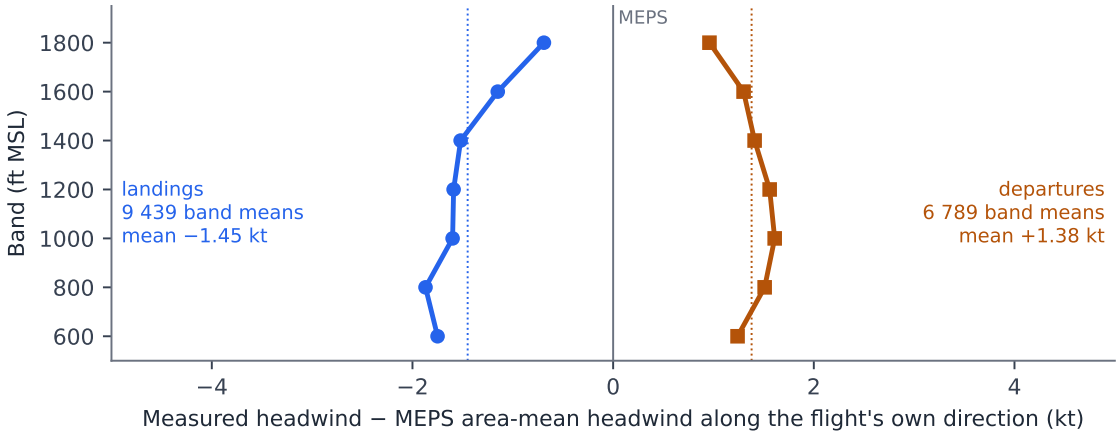


Figure 6. Headwind of landings and departures minus the MEPS area-mean headwind along each flight's own direction, by 200-ft band. The dotted lines are the means used as the phase correction.

The most likely explanation is an airspeed effect of the flight phase — the reported TAS of an aircraft in approach configuration (flaps and gear out, low speed, high angle of attack) is slightly low and that of a climbing aircraft slightly high, a pressure-error / calibration effect of the air-data system — rather than a real wind difference. It does not matter for the ramps, because the offset is constant within one flight and cancels in ΔV. It does matter for the band comparison, which mixes landings and departures.

6.3 Change

A band is compared with **references of the same phase** (landings with landings, departures with departures) when it has at least 5 of them. Otherwise all references are used, but every vector — the references' and the flight's own — is first corrected by the phase bias b along its own direction:

$$\mathbf{w}' = \mathbf{w} + b_{\text{phase}}\,\hat{\mathbf{d}}, \qquad b_{\text{arr}} = -1.45\,kt, \quad b_{\text{dep}} = +1.38\,kt. \tag{4}$$

The report records which was used (`ph`: same / all). A first trial with 3 same-phase references produced a spurious landing WSA on 17 October at 09:46, so 5 was chosen.

> **Why the change.** At night there are often only a few landings, and departures are the only recent references; mixing them is valuable, but only after the systematic phase difference has been removed. Same-phase references need no correction at all, so they are preferred whenever there are enough of them.

Table 6. Replay of v4 and v5 on all stored flights, 9–27 October 2026.

	v4	v5
Landings: band WSA	4	4
Landings: mean band change	−0.79 kt	+0.21 kt
Departures: mean band change	+1.57 kt	−0.33 kt
Departures: band ADV + WSA	27 + 9	28 + 7
Departures: flags with more / less headwind	27 / 9	19 / 16
Departure ADVs 26–27 October	5	2
17 October 09:34 departure	WSA 20.2 kt	ADV 12.2 kt

The departure flags are now balanced between more and less headwind, as they should be if they come from the weather and not from the method.

7 Phase 5 — higher band limits for turboprop departures, v6 (27 October 2026, evening)

7.1 Observation: ATR departures flag five times as often

In the evening data check of 27 October [20] all three departure advisories of the day were ATR 72 climb-outs (FIN4BX, FIN2PE, FIN6WR), each with 11–15 kt *less* headwind at 1 600–1 800 ft. Over all stored departures, 20 of 688 turboprop

climb-outs (2.9 %) had a band advisory or alert, against 18 of 2 946 jet climb-outs (0.6 %); the turboprop flags were mostly "less headwind" (32 of 49 changes of at least 8 kt), and largest at 1 400–1 600 ft. The landings of the same turboprops showed nothing unusual (0.3 %).

7.2 Tests

The question was whether the cause is an *offset* of the aircraft type (to be subtracted) or a larger *scatter* (to be allowed for). The band comparison was replayed for every flight of 13–27 October with the change of *every* band, not only the strongest one (the strongest band of each flight gives a biased picture, because it is selected for its size).

- **Per-type offset:** real but small, and repeatable. Learnt on 13–18 October and compared with 19–27 October, the type means agree ($r = 0.90$ over 33 type–phase pairs): E190 +1.3 / +1.4 kt (departures / landings), A220 +1.3 / +1.2, A330-300 −0.7 / −1.4, A321 −0.3 / −1.1, ATR 72 −0.9 / −0.2 kt. Removing them (test period) narrowed the scatter of the jets by about a tenth (landings 2.12 → 1.92 kt, departures 2.64 → 2.51 kt), but hardly that of the turboprop departures (3.26 → 3.23 kt); their flags fell only from 3.3 to 2.5 %.
- **Scatter:** the turboprop climb-outs scatter more (Figure 7) — SD 3.2–3.8 kt at 1 200–1 800 ft against 2.7–2.9 kt for the jet departures — although they have three to seven wind pairs per band (they climb slowly). The heights are where an ATR accelerates, retracts its flaps and makes its first turn; the band means there are also 1.0–1.6 kt low. The ratio of the scatters is about 1.3.

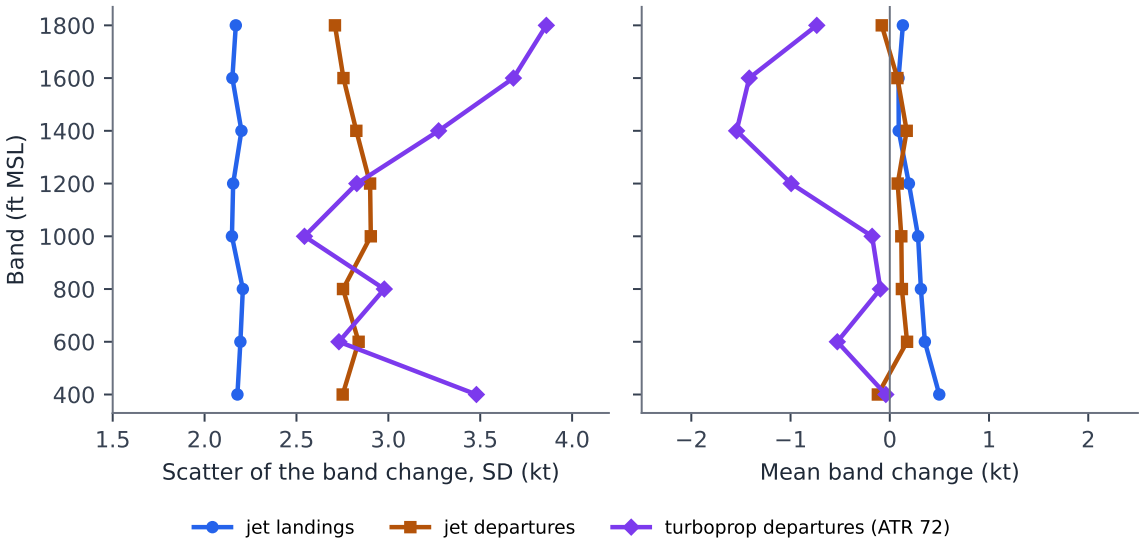


Figure 7. Band change of each flight against the flights of the previous 45 minutes, every band, 13–27 October 2026 (v5): scatter (left) and mean (right) for jet landings (20 136 band changes), jet departures (14 713) and turboprop departures (2 879, almost all ATR 72).

7.3 Change

For the departure of a turboprop (ICAO types ATR 42/72, Dash 8, Saab 340/2000, Fokker 50, Do 328, Jetstream, Beech 1900 / King Air, PC-12, Caravan, L 410, Brasilia, Metro) the band limits are multiplied by `TP_DEP_SCALE` = 1.3: **advisory 13 kt, windshear alert 19.5 kt**, also in the rule for a band resting on one wind pair. Jets and all landings keep 10 / 15 kt. The report records the limit used (`thr`), and the hover of the alert and of the ACI + WS chart shows it.

Why the change. A limit should stand at the same distance from the normal variation for every aircraft. The jets' 10 kt is about 3.5–4 times the scatter of their climb-outs; 13 kt keeps the ATR at the same distance. The per-type offset is kept in reserve: it is too small to matter for the alerts, but may be worth removing later for the accuracy of the band winds themselves.

Table 7. Replay of v5 and v6 on all stored flights, 9 / 11 – 27 October 2026 18 UTC.

	v5	v6
Turboprop departures (711): band ADV + WSA	18 + 3	8 + 1
Jet departures (2 987): band ADV + WSA	13 + 4	13 + 4
Landings (3 913): all levels	—	unchanged
27 October: FIN6WR (−10.7 kt) / FIN4BX (−14.2) / FIN2PE (−14.9)	ADV / ADV / ADV	none / ADV / ADV

Of the 14 turboprop departures whose level changed, 12 went from ADV to none (changes of 10.1–11.9 kt) and 2 from WSA to ADV (16.5 and 19.3 kt). The departures now flag at a similar rate whatever the type: 1.3 % of the turboprops, 0.6 % of the jets.

8 New tools for the study (27 October 2026, late evening)

Two additions that change nothing in the alerts but give the study new material.

8.1 Space–time diagram of the band changes

The detection stores for each flight only its *largest* band change. To see how a change travels through successive flights, the change of *every* band of every landing and departure is now recomputed on request from the stored band profiles, exactly as in the band comparison (same references, phase correction and sign), and drawn as a time–height picture: the **ST** view of the Windshear page. Each cell is the median of the flights in a 5–30-minute slot. A front or a shear layer is a coloured streak through successive flights; the eddy of a single aircraft is damped by the others in its slot. Figure 8 shows the wind change of 17 October: the whole layer from 600 to 1 800 ft lost headwind at 09:00 (the landings FIN2EZ, RYR9857, FIN2AT and the departure FIN863 before them), and the next slots show the reversal after the runway change. On the turbulent 27 October the same diagram has only scattered single cells.

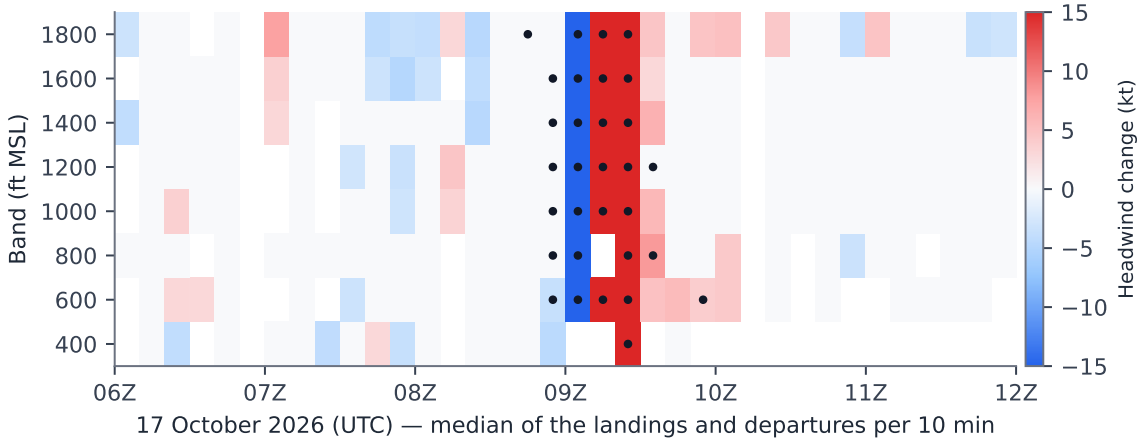


Figure 8. Space–time diagram, 17 October 2026 06–12 UTC: the median headwind change per 10 minutes and 200-ft band of the landings and departures, each against the flights of the previous 45 minutes (blue less, red more headwind; changes below 3 kt white). Dots: a flight at or beyond its advisory limit.

8.2 Climb-out series and trial ramps along the departures

Until now the departures stored only their band profile, so the ramp analysis could not be applied to them. From 27 October every climb-out keeps its reply series (BDS 5,0 / 6,0 replies with the height at the reply time, the ADS-B velocity messages and the sweeps; table `departure_series`, the format of the approach series), and at the end of the climb the ramp analysis of the landings is run on it — the low-level zone being the first 1 600 ft above the airport or 5 NM from it. The result is stored in the departure report as `trial`, with the level it *would* have; **no alert is raised**. Before the ramps may alert, the stored climb-outs must show how the acceleration (speed changes and flap retraction in the first 1 500 ft) affects TAS − GS, and whether the 15–25 replies of a fast climb in the zone (about 100 on final) are enough. Planned check: after one to two weeks, compare the trial ramps of successive departures and with the landings' ramps and band changes.

9 The first night of the departure trial: a nocturnal low-level jet (28 October 2026)

9.1 Observation

In the night of 27–28 October a south-westerly flow blew over a stable surface layer: the Jokioinen sounding of 00 UTC had 12 kt at the ground, 21 kt at 1 050 ft, 34 kt at 2 000 ft and 43 kt at 3 100 ft (MSL) — a *nocturnal low-level jet*, the wind above the inversion running free of the ground friction. EFHK landed on runway 15 and took off from 22L/22R (track about 228°), so every climb-out flew straight into the jet: its headwind grew by about 12 kt per 1 000 ft (Figure 9a). For the landings on 15 the same wind came 60–70° from the side, so only a third to a half of it was headwind; no landing was flagged that night.

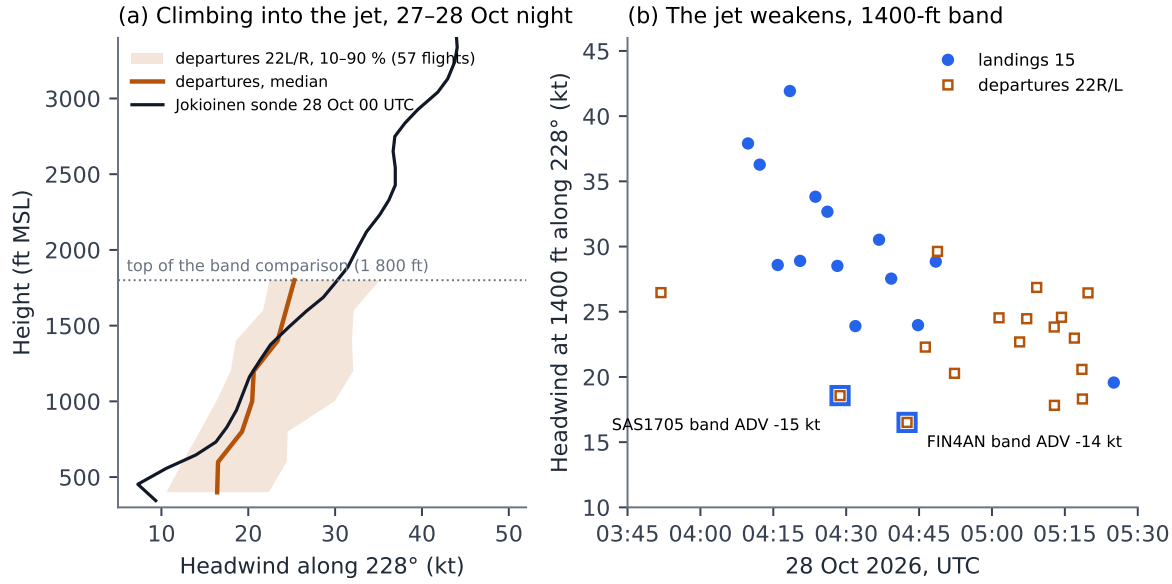


Figure 9. (a) Headwind along 228° of the departures from 22L/22R, 27 October 21 UTC – 28 October 06 UTC (band profiles, phase bias removed; median and 10–90 % range), and of the Jokioinen sounding of 28 October 00 UTC. (b) Headwind in the 1 400-ft band, 28 October 03:45–05:30 UTC: landings on 15 and departures; boxed: the two departure band advisories.

9.2 The trial ramps flag the jet itself

Of the 61 departures between the start of the trial (27 October 19:40 UTC) and 06 UTC, 57 got a trial ramp analysis (4 had too few replies), and 4 would have flagged — 7 %, against about 1 % of the landings (Table 8).

Table 8. The trial ramps that would have flagged, 27–28 October night, with the change of the sounding's headwind (along 228°) over the same heights.

Flight	Type	Trial	Ramp	Heights (ft)	Sounding over the same heights
FIN963	E190	ADV	gain 18.7 kt (14.8 per km)	492–2 119	+25.1 kt — the jet
BTI32M	BCS3	ADV	gain 13.0 kt (11.4 per km)	710–2 118	+17.9 kt — the jet
FIN4B	E190	WSA	gain 14.0 kt (15.1 per km)	907–1 522	+7.0 kt — twice the profile over 600 ft
FIN375	AT75	ADV	loss 16.3 kt (10.3 per km)	776–1 755	+12.8 kt (a gain) — the turboprop climb pattern of Phase 5

Two of the four are the vertical wind profile, measured correctly: the climb-out met *less* increase of headwind than the sounding had over the same heights. A landing ramp of that size would be judged against the other flights' profile (the relative rule of Phase 2), but here that check could not run: the ramps end at 1 970–2 120 ft, while the reference profile of the band comparison stops at 1 800 ft; and two ramps are at or above the absolute limit of 12 kt per km, where the relative rule does not apply. FIN4B (WSA) had twice the profile's change in a short ramp — possibly real, but on one aircraft. FIN375 shows the ATR 72's known loss of airspeed-based headwind during flap retraction (Section 7), against a profile that was gaining.

9.3 Two departure advisories during the weakening of the jet

At 04:28 and 04:42 UTC two departures got a band advisory at 1 400 ft: SAS1705 (A20N, −14.8 kt) and FIN4AN (A321, −13.9 kt). Figure 9b shows why: the landings at 1 400 ft went from 36–42 kt (04:09–04:18) to 24–28 kt (04:31–04:48) — the jet was weakening — and the departures read about 5 kt lower still, with only 2–3 wind pairs per band. With fewer than five departures in the previous 45 minutes they were compared with all flights (phase bias removed), mostly the earlier landings. The sign and size of the change were right; this is the band comparison working as designed during a change in time, sharpened by the spatial difference between the approach to 15 (north-west of the airport) and the climb-out from 22 (south-west), which reached 5–7 kt at 1 200–1 600 ft in this hour against about 1.5 kt during the rest of the night. The live alert of SAS1705 was a WSA (−15 kt against 8 references); the stored report, with 9 references, an ADV.

9.4 What this means for the trial

A stable night with a low-level jet is not a rare situation, so the departure ramps need an expectation before they may alert, as the landings got in Phase 2. Two questions for the evaluation (Section 19): **(1)** extend the reference profile above the low-level zone (the departures' bands up to about 3 000 ft) or cut the ramps at the top of the zone; **(2)** whether a headwind *gain* on a climb-out should flag at all — it increases the aircraft's performance, but ICAO and LLWAS alert on gains as well.

10 Comparison with the Hong Kong lidar method (28 October 2026, evening)

The ramp method of Phase 1 was taken from the Hong Kong Observatory (HKO) lidar algorithm GLYGA [4, 6]. On 28 October the two lidar papers were read in full, together with two HKO papers to the ICAO Asia/Pacific meteorology group [22, 23], and three things were checked against the stored landings: GLYGA's alert rule, its "gentle ramp" finding and its alert-duration measure. Nothing in the detection was changed; the results are the basis for a decision.

10.1 What the papers say

- **GLYGA** [4, 6]: a headwind profile on a 100-m grid along each glide path from the lidar's glide-path scans; spikes replaced by a median value; ramps searched at lengths 400, 800, …, 6 400 m on a smoothed increment profile, then widened or narrowed to the full change. Overlapping ramps are ranked by the severity factor of Woodfield and Woods (1983, as cited there), $S = (\Delta V / H^{1/3})^3 / V_{\mathrm{app}}$, so the ranking depends on $\Delta V / H^{1/3}$. **The alert itself is raised on the headwind change**: $|\Delta V| \geq 14\,kt$ (the 15-kt standard, lowered by 1 kt because the beam sees only part of the headwind). The $H^{1/3}$ scaling only chooses which ramp is reported.
- **Performance** [4]: on the most-used approach corridor in spring 2006–07, 76 % of the pilot reports of significant windshear were covered by an alert, with a false-alarm rate of 34 %. Pilots perceive shear differently (10 and 20 kt reported for similar lidar ramps a few hours apart), and flight-data (QAR) checks of eight "false alarms" found significant headwind changes in most of them. In a Woodfield diagram (severity against ramp length) the false alarms gathered at ramps longer than 3 km.
- **Gentle ramp removal** [6]: an alert is withheld when its ramp is longer than 3 000 m and the profile fluctuates less than 1.2 kt RMS about its running mean. Over 46 months (2016–19, 1 557 quality-controlled pilot reports) this cut the alert duration on the main corridor from 2.14 to 1.26 % of the time, while the hit rate fell only from 61.3 to 57.4 %. GLYGA typically reaches a hit rate of 50–70 % with an alert in force 2–4 % of the time.
- **Mode S at HKO** [22]: winds and temperatures are derived from the Shenzhen radar's Mode S data (BDS 5,0 and 6,0) with positions from ADS-B, are comparable to AMDAR, and feed the approach-spacing system of Hong Kong's air traffic control. Windshear alerting is named only as a suitable application; **no windshear detection from Mode S is described**. The radar interrogates each aircraft every 4 s and gets BDS 5,0 and 6,0 together only once in about 24 s. The headwind of this study needs only BDS 5,0 (TAS and GS of the same reply), which the Finnish wide-area multilateration system brings on final every 2 s (median 1.97 s, Section 1) — one of the reasons a headwind profile per aircraft is possible here.
- **Validation** [23]: HKO verifies its windshear alerting against pilot reports, relayed by air traffic control, and uses QAR data after the event. Neither is available to this study; its proxies remain the confirmation by the neighbouring landings (Phase 3), go-arounds and METAR windshear remarks.

10.2 Replay of the 14-kt rule

All landings with stored raw replies (9 October 11 UTC – 28 October 19 UTC; 4 155 landings, 4 097 with a profile) were analysed again with the current detector and with one extra condition: a ramp may reach WSA (or MBA) only if its headwind change itself is at least 14 kt; the advisory level is unchanged. The replay reproduces the stored ramps exactly (0 differences in 8 194 loss and gain ramps).

Table 9. Ramp levels of the landings, 9–28 October 2026, now and with $|\Delta V| \geq 14\,kt$ for WSA. Limits of 12, 13 and 15 kt give the same result.

Ramp level	none	ADV	WSA	MBA
Now (v6)	4 011	83	3	0
WSA only with $\lvert\Delta V\rvert \geq 14\,kt$	4 011	85	1	0

Table 10. The three ramp WSAs of the period (all on 26 October, runway 15).

Time	Flight	Type	Ramp	Heights (ft)	With the 14-kt rule
14:06	FIN7SM	A319	loss 11.3 kt over 400 m (15.4 per km)	1 123–1 048	ADV; confirmed by 2 earlier landings
17:45	FIN9LK	E190	gain 15.0 kt over 1 000 m (15.0 per km)	825–646	stays WSA; isolated (0 of 5 before, 0 of 3 after)
21:06	FIN9EJ	AT75	loss 11.2 kt over 400 m (15.2 per km)	1 393–1 337	ADV; isolated

Two of the three WSAs were changes of only 11 kt, raised to 15 kt per km by the scaling of a 400-m ramp ($(1\,\mathrm{km}/0.4\,\mathrm{km})^{1/3} = 1.36$). A 400-m ramp is measured by three or four replies at each end; neither change reaches the 15-kt definition of significant windshear in absolute terms. The station's own alert log of 25–28 October agrees: the same three ramp WSAs, plus one band-comparison WSA (SAS1705, the departure of Section 9).

Proposal (v7, not yet implemented): WSA and MBA also require $|\Delta V| \geq 14\,kt$, as in GLYGA; the 1-km equivalent still ranks the ramps and sets the advisory. The cost seen in this period is one confirmed ramp (FIN7SM) going from WSA to ADV. *(Later note: implemented the same evening as v7, Phase 6, Section 11.)*

10.3 Severity against ramp length (Woodfield diagram)

Figure 10 shows the strongest loss and gain ramp in the zone of every landing (1 563 ramps of 1 465 landings; most landings have only weak ramps, below the plot's significance test). The two level curves are curves of constant severity factor: with $V_{\mathrm{app}} = 75\,\mathrm{m\,s^{-1}}$, 10 kt per km is $S = 1.8 \times 10^{-3}$ and 15 kt per km $S = 6.1 \times 10^{-3}\,\mathrm{m\,s^{-2}}$.

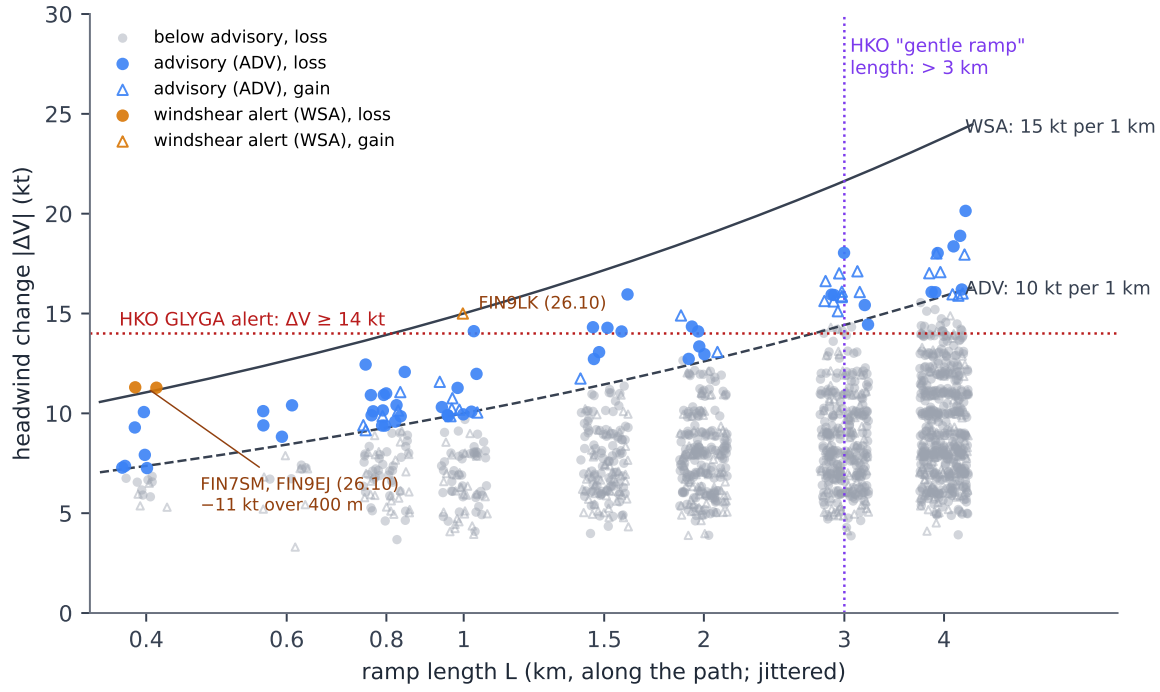


Figure 10. Headwind change against ramp length for the strongest loss and gain ramp in the zone of each landing, 9–28 October 2026, with the ramp level before the relative rule of Phase 2. Curves: the ADV and WSA levels (constant 1-km equivalent = constant severity factor); red: GLYGA's 14-kt alert; violet: HKO's 3-km "gentle ramp" length.

- **Long ramps:** 71 ramps of 3–4 km had a change of 14 kt or more. GLYGA would have alerted on all of them (before its gentle ramp removal); here 42 stayed below the advisory level and 29 were advisories, none a WSA. The $L^{1/3}$ scaling already does at Helsinki-Vantaa what HKO's gentle ramp removal does at Hong Kong — in a stable night the normal increase of the wind with height is exactly such a ramp (Section 9).
- **Short ramps:** below 1 km the scaling works the other way and lifts 10–11-kt changes to the WSA curve (Table 10). This is where the 14-kt rule acts.

10.4 Alert duration

HKO describes its alerting by the share of time an alert is in force on a corridor. The stored reports give an estimate for MODE-S Wind: each flagged landing or departure is taken to hold an alert from 2 min before its landing or take-off time

to 10 min after it (the live alert stays 10 min after it was last seen), merged per runway direction (Figure 11).

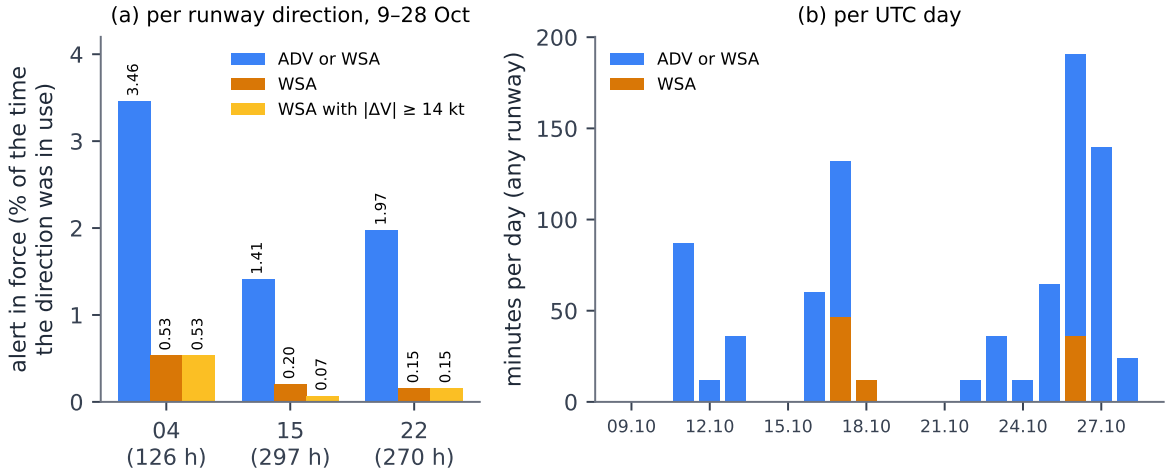


Figure 11. Estimated alert duration, 9–28 October 2026 (landings and departures, ramp and band levels). (a) Per runway direction as a share of the time it was in use (half-hours with a landing or departure on it); the third bar is the WSA duration with the 14-kt rule. (b) Minutes per UTC day with an alert on any runway.

- **ADV or WSA:** 2.9 % of all time on any runway; per direction 3.5 % (04), 1.4 % (15) and 2.0 % (22) of the time it was in use.
- **WSA:** 0.34 % of all time (0.53, 0.20 and 0.15 % of the in-use time of 04, 15 and 22); with the 14-kt rule 0.25 % (runway 15: 0.07 %).
- The alerts come in a few days — the 17 October wind change and the windy 25–27 October; most days have none.

GLYGA's single alert level corresponds to the WSA here. Its 2–4 % is about ten times the WSA duration at Helsinki-Vantaa, at an airport where about one flight in 500 reports significant windshear [4]. Without pilot reports this is not a comparison of performance, but it shows that the alerts here are rare and short, as an alerting service needs them to be.

11 Phase 6 — a windshear alert needs a 14-kt change, v7 (28 October 2026, late evening)

Why the change. The 1-km equivalent ranks ramps well, but at the short end it lifts small changes: a 400-m ramp counts 1.36 times its size, and it rests on three or four replies at each end and on the 400-m running mean of the profile. Two of the three ramp windshear alerts of the study were changes of 11 kt (Table 10), below the 15-kt definition of significant windshear. The HKO lidar alerting, from which the ramp method comes, alerts on the change itself (Section 10).

The change. A ramp reaches WSA, or MBA, only when its headwind change is at least 14 kt (`RAMP_WSA_MIN_DV`); the 1-km equivalent still ranks the ramps and sets the advisory. Of two ramps of an approach, the one that sets the level is the one reported. Only ramps shorter than 1 km can be affected, since from 1 km up the 1-km equivalent is never larger than the change. The live alerts, the stored reports and the trial ramps along the departures use the same rule; at the first start the stored reports are re-levelled (report version 7).

14 rather than 15 kt. The headwind here is measured along the aircraft's own track, so the reason HKO lowered 15 to 14 kt (the lidar sees only part of the headwind) does not apply. But the 400-m running mean flattens a short, sharp change a little, and 14 kt leaves that margin. In this period 12, 13, 14 and 15 kt gave the same result.

Effect (replay of Section 10, 9–28 October 2026, 4 155 landings): ramp WSAs 3 → 1, ramp advisories 83 → 85, no other change. FIN7SM and FIN9EJ (26 October, 11-kt losses over 400 m) are advisories; FIN9LK (a 15-kt gain over 1 km) stays a WSA. The estimated WSA time falls from 0.34 to 0.25 % (Figure 11). The confirmation label is unchanged. To be reviewed in the first convective season, when gust fronts and microbursts give short, strong ramps.

12 Turbulence (EDR) from the replies (29 October 2026)

Why the change. The isolated ramp alerts of the turbulent days (Phase 3) are eddies, not shear layers, and the Approach Conditions Index (ACI) that describes the roughness of an approach depends on the aircraft class (class factors 1.0–1.5). A measure of the turbulence of the *air*, in the units ICAO uses, would tell turbulence and windshear apart and could be compared with pilot reports.

12.1 Method

The ICAO turbulence measure is the eddy dissipation rate EDR $= \varepsilon^{1/3}$ ($\mathrm{m}^{2/3}\,\mathrm{s}^{-1}$); Annex 3 [1] grades the *peak* EDR of a medium aircraft as light from 0.10, moderate from 0.20 and severe from 0.45. It was estimated from the headwind TAS − GS of the BDS 5,0 replies, after the structure-function approach of Kopeć et al. [27].

A first version was rejected. A 60-s running-mean detrend with a fit $D = A\,r^{2/3} + N$ over 100–1 600 m correlated well with the ACI day by day (0.83), but most of the signal came from *turns*: in a turn the headwind part of the mean wind changes. On straight segments only, the daily correlation fell to 0.26. The structure function of the straight segments is also nearly flat (slope 0.14, not $2/3$): replies about 2 s (∼140 m) apart do not resolve the inertial range. On turbulent days the whole curve moves up ($D(141\,\mathrm{m})$ 0.4 on calm days, 1.4–1.8 $\mathrm{m}^2\,\mathrm{s}^{-2}$ on 16 and 25–27 October).

Adopted (v1 of `collector/edr.py`). The replies 200–3 000 ft above the threshold are cut into straight pieces of about 2 km (track change $\leq 15°$, ≥ 8 replies), each linearly detrended; the mean squared headwind difference of all reply pairs 100–200 m apart, D, minus the noise of the 2-kt TAS / GS resolution $N_0 = 0.35\,\mathrm{m}^2\,\mathrm{s}^{-2}$ (the floor of the calmest day), gives

$$\mathrm{EDR} = \sqrt{\frac{D - N_0}{2 \cdot 141^{2/3}}}$$

with the inertial-range law $D = 2\,\varepsilon^{2/3} r^{2/3}$ *assumed* at $r = 141$ m. Per flight: the pooled value over all pieces ("EDR"), 200–1 000 and 1 000–3 000 ft, the strongest piece and the number of pieces. Computed when a landing or departure is stored; the stored series are back-filled at start.

12.2 Results (snapshot 29 October 2026; 4 255 landings 9–29 October)

- **Distribution:** 5 / 25 / 50 / 75 / 90 / 95 / 99 % = 0.00 / 0.031 / 0.069 / 0.114 / 0.156 / 0.181 / 0.231. 29 % of the landings reach 0.10 and 2.4 % 0.20 with the averaged value; with the strongest piece 53 and 14 %.
- **Against the ACI:** Spearman 0.66 per landing, daily medians 0.98 (21 days); Pearson 0.67 on 3 293 landings with both (Figure 12). The median EDR reaches 0.10 at an ACI of about 3.3 and 0.15 at about 5.
- **Against the surface wind** (Spearman 0.68): median 0.028 at 0–4 kt, 0.046 at 5–9, 0.101 at 10–14, 0.156 at 15–19 kt. Larger below 1 000 ft (0.092) than above (0.058): mechanical turbulence of the surface layer. Consecutive landings within 5 minutes correlate at 0.61.
- **Aircraft class:** the ratio to the neighbouring flights (±45 min) is 1.01 (regional), 1.04 (narrow-body), 0.98 (wide-body), 1.06 (business) and 0.85 (turboprop); the correlation with the ACI is 0.67–0.69 in every class. The EDR describes the air; it needs no class factors.
- **The windshear flags come in turbulent air:** the landings with a confirmed ramp alert had a median EDR of 0.145, the isolated ones 0.16, the landings without a level 0.065 (25 / 12 / 3 253 flights). The isolated alerts are slightly rougher still — consistent with Phase 3, small samples.

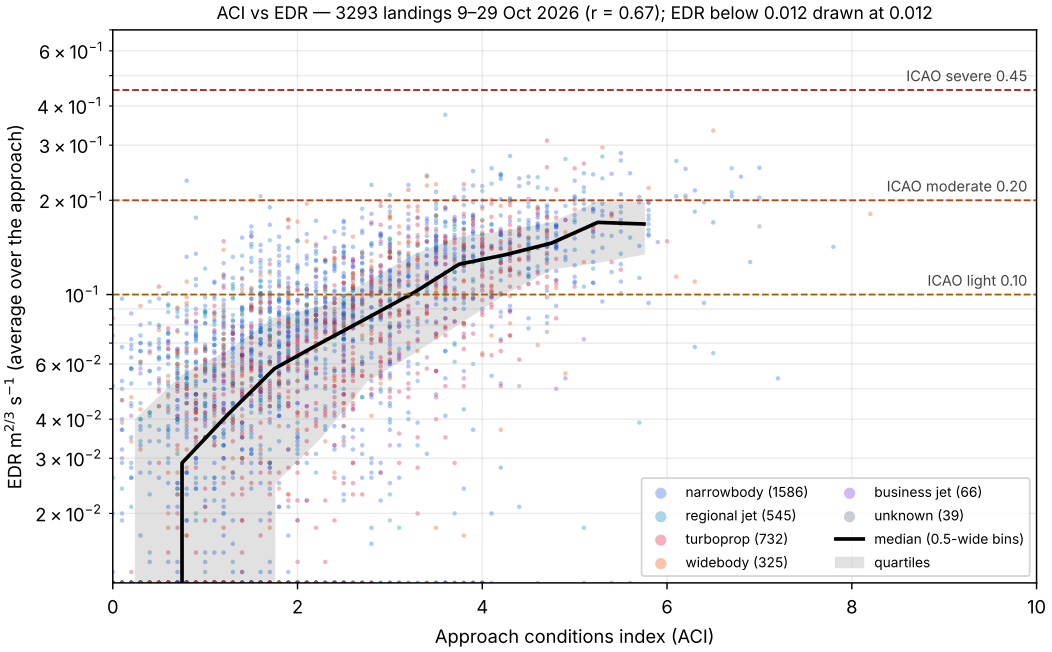


Figure 12. ACI against EDR for 3 293 landings, 9–29 October 2026, by aircraft class (logarithmic EDR axis; values below 0.012 drawn at 0.012). Black: the median in ACI bins of 0.5 with the quartiles; dashed: the ICAO limits for the peak EDR.

12.3 Departures: why the series now reaches 5 000 ft

Only 34 % of the climb-outs got a value (landings 78 %): a climb crosses 200–3 000 ft in about 80 s with a median of 37 BDS 5,0 replies, and it turns early, so half of them had no straight 2-km piece. Two ways to use the turning parts were tried on the stored series (361 climb-outs, with 1 500 landings as the check):

- **the wind vector** from TAS and the BDS 6,0 heading (interpolated to the reply time, 10.5° declination) against GS and track: coverage 34 → 49 %, but the values agreed with the straight-piece headwind EDR only at 0.28 on departures (0.93 on landings);
- **turning pieces** (15–60°) with a quadratic detrend: coverage 34 → 48 %, agreement 0.33 (landings 0.48), and lower values — the detrend removes part of the turbulence.

Neither was adopted. Instead, from 29 October the climb-out series is stored up to 5 300 ft (the bands, the band comparison, the trial ramps and the windshear report still end at 3 300 ft as before), and the departures' EDR uses 200–5 000 ft — about 1 kB more per departure. The coverage is to be checked after a week of data.

12.4 Shown and open

The EDR is shown with the ACI and the windshear on one time axis, in the Approach History (EDR view, with a filter ≥ light / ≥ moderate), on the phone, and as coloured pieces of the paths on the phone's approach map. Open: the calibration. The value is an average over the approach from replies 2 s apart, not the ICAO one-minute peak; the ICAO lines are a guide only until the stored turbulence reports (Section 14) give enough cases.

13 The boundary layer: low-level jet and surface inversion (29 October 2026)

Why the change. The first night of the departure trial (Section 9) showed that the vertical wind profile of a stable night alone can give windshear flags. The detection needs to know when the boundary layer is stable, and when a low-level jet is present.

13.1 Method (`collector/boundary.py`)

Every hour: the median wind (u, v) of all aircraft within 25 km of the airport in 250-ft bins from 250 to 5 000 ft MSL (pressure altitude corrected with the QNH; ≥ 3 observations per bin), and the temperatures of the airport low-level profile (200-ft bands 400–3 000 ft from the landings and departures, ≥ 4 flights, smoothed over ±200 ft) against the METAR temperature.

- **Low-level jet:** the criterion of Baas et al. [28] (a maximum at least 2 m/s and 25 % above the minimum above it) as such flagged 39 % of the hours — it also catches the broad decrease of wind with height. Made stricter: a local

maximum (the "nose") at most 2 000 ft above the airport whose speed falls by at least 2.5 m/s and 25 % *within 1 500 ft* above it, and at least 2 m/s above the lowest speed below it.
- **Surface inversion:** the air at 600–1 000 ft at least 0.5 K warmer than the METAR temperature; the top is the highest band still at least half that much warmer.

Check of the aircraft temperatures: in windy daytime hours ($\geq$ 12 kt) the 600–1 000-ft air was 1.75 K colder than the METAR — the dry-adiabatic 1.8 K over 190 m — so the aircraft read the low-level temperature without a measurable bias.

13.2 Results (29 September – 29 October 2026)

- **Low-level jet** in 10 % of the hours: 16 % at night (18–06 local), 4 % at midday; typically at 1 500 ft and 23 kt, 3.5 m/s weaker above, about 6 kt per 1 000 ft of shear below. Most jet hours on 4, 11, 12 (easterly flow), 18 and 23 October.
- **Surface inversion** in 15 % of the 367 hours with a METAR temperature: 23 % at night, 2 % at midday; in 49 % of the hours with a wind under 5 kt, 24 % at 5–10 kt, never from 10 kt; 21 % in clear and 5 % in cloudy hours. Strongest on the clear, calm night of 19–20 October, +6 to +9 K at 600–1 000 ft. Jet hours were not more often inverted (17 against 15 %).
- **The night of 27–28 October** (Section 9) is *not* a jet by this criterion: the wind kept increasing to 3 000 ft and more (the sounding: 43 kt at 3 100 ft), so there is no nose; it was the strong shear of the stable layer under a free flow. The strict criterion did find a south-westerly jet the next night (28 October 21–22 UTC and 29 October 04–08 UTC, 19–25 kt at 1 500–2 000 ft).
- **Gap:** between 01 and 05 local there are few flights, so the peak hours of the night jet are often unseen.



Figure 13. The boundary layer, 17–19 October 2026 (Sounding page, time–height section, field *Boundary layer · wind*): the wind speed in 250-ft steps, the low-level jet (◊ with its speed; the north-easterly jet of the evening of 17 October behind the cold front, 30→24 kt) and the surface inversions under the high (red bar, red line at the top). Hatched: no flights.

13.3 Relation to the windshear flags and the turbulence

Of 4 261 landings 9–29 October with a boundary-layer hour:

- **Windshear levels** were *rarer* during a jet (2 of 421 landings, 0.5 %) than without one (47 of 3 840, 1.2 %), and absent during a surface inversion (0 of 460 against 49 of 3 801). The median band change was 2.6 kt in jet hours and 2.9 kt otherwise. In this autumn sample the flags come with the windy, turbulent days, not with the stable nights; the jet of 27–28 October gave departure flags (Section 9) and none on landings, because the landings on runway 15 met it from the side.
- **Turbulence:** median EDR 0.048 in jet hours, 0.019 under an inversion, 0.076 with neither. At the same surface wind the low-layer EDR was lower in jet hours (5–9 kt: 0.027 against 0.054) and nil under an inversion (0.00 against 0.025–0.058): the stable layer decouples the surface and damps the turbulence.

Since 29 October each windshear alert and report shows the boundary layer of its hour (an `LLJ` / `INV` tag and a line "Boundary layer then: …"), and the Skew-T airport profile marks the jet and the inversion, so that each future flag can be read against it. The first case: the advisory of FIN440 on 04L on 23 October 2026 14:15 (a 17-kt headwind gain over 4 km at 2 240–1 580 ft) came in an hour with a south-easterly jet of 27 kt at 1 750 ft — mostly across the runway, so it explains the gain only in part.

Sea breeze (spring and summer): a METAR wind turning onshore (south) with a temperature drop by day and the lowest band winds turning first; to be added to the boundary-layer hours in April 2027.

14 Validation data: METAR archive and pilot reports (29 October 2026)

- **METARs:** every METAR of EFHK is stored with its parsed values and its **windshear group** (`WS R22L`, `WS ALL RWY`). In Finland the group is given only in manual METARs, for wind shear between the surface and 1 600 ft observed during a landing or take-off; there are no SPECI (half-hourly METARs), and a local SPECIAL goes to the ATIS only [29]. A loader brings the history from the Iowa Environmental Mesonet archive since 2018 (one request per month), so the days with a windshear group can be compared with the stored flights.
- **Pilot reports:** the ARS and the national WXREP (reports below the ARS criteria: light turbulence, light icing, light windshear, *inversion*, CB tops) and the SIGMETs of the FMI aviation weather service are fetched every 30 minutes; one report becomes an ARS or a WXREP, never both. Reports within 30 NM of the airport are marked "near". The first day brought one report (moderate icing at FL140–160, 150 km away).
- **Plan:** score the alerts against the windshear METARs and the windshear pilot reports per runway and hour (hit, miss, false alarm), as HKO does [23]; and match each turbulence report near the airport with the EDR of the flights within 30 minutes — the calibration of Section 12. This needs weeks to months of reports.

15 Phase 7 — the ADS-B ground speed and the BDS 6,0 airspeed, v8 in shadow (30 October 2026)

15.1 Observation

The headwind of v2–v7 comes from one register: TAS − GS of each BDS 5,0 reply, about one every 2 s on final, both speeds in 2-kt steps. Two other sources reach the receiver at the same time: the **ADS-B airborne velocity** (TC 19), broadcast by the aircraft itself about twice a second with the ground speed in 1-kt steps — stored with every landing since 24 October 2026 and every departure since 27 October — and the **BDS 6,0** register with the indicated airspeed (1-kt steps) and the Mach number, answered about 1.5 times as often as BDS 5,0. The indicated airspeed becomes a true airspeed with the pressure altitude and the temperature. Together they could give the headwind several times as often, which matters most for the short ramps of 400–600 m that the replies of BDS 5,0 alone resolve poorly.

15.2 Tests

All tests on the snapshot of 30 October 2026 (09:12 UTC): the 1 062 landings of 24 October 20:41 – 30 October 09:00 UTC with ADS-B ground speed and BDS 5,0 replies.

- **Timing.** The ADS-B ground speed of 0.5 s *before* a reply matches the reply best: BDS 5,0 GS minus ADS-B GS has an RMS of 0.92 kt (bias +0.3 kt), mostly the 2-kt steps of the register; at the reply's own time 0.94 kt, 1 s later 1.21 kt.
- **Temperature for BDS 6,0.** The flight's own deviation from the ISA, the median over the pairs of a BDS 5,0 TAS and a BDS 6,0 Mach within 1 s ($T = (\mathrm{TAS}/38.97\,M)^2$), was −5.2 K on the median landing; turboprops send no Mach, so 20 % of the flights take the median of the flights within ±1 h. A 1-K error is 0.17 % of the TAS, 0.24 kt at 140 kt — and it changes only slowly along an approach.
- **Agreement of the two registers.** The headwind from the BDS 6,0 airspeed against the ADS-B ground speed, compared with that from BDS 5,0 against the same ground speed at the same moment: bias 0.02 kt, RMS 0.98 kt. Per flight the offset between them is small (p10 / p90 −0.43 / +0.41 kt) and is removed, so that a change in the mix of the two registers along the path cannot read as a ramp.
- **Noise and density.** About the ±7.5-s running median: 1.53 kt for TAS − GS of BDS 5,0, 1.44 kt for BDS 5,0 TAS against the ADS-B GS, 1.42 kt for BDS 6,0, 1.44 kt combined. A value every 0.52 s instead of 1.94 s: 135 values per approach accepted by the analysis instead of 59 (2.3×), so the 400-m means are about 1.5 times less noisy.
- **A trap.** The repeat filter (the same register content read again within 6 s) must look at the register's own GS and TAS; with the ADS-B GS substituted first, stale BDS 5,0 contents slip back in with a fresh ground speed.
- **Replay of the ramps** (Table 11). With the rules of v7 (advisories of 10–12 kt per km relative to the reference flights), the stored-report replay of 24–30 October gives 41 ramp advisories with v8 against 30 advisories and 1 WSA with v7; 23 are new, 14 drop out, 17 are flagged by both. The new ones are mostly short: 24 of the 41 v8 advisories have L = 400 m. The one WSA (26 October 17:45 UTC, an E190 on runway 15, exactly 15.0 kt per km) reads 13.7 kt and becomes an advisory.
- **Are the new ramps real?** Each advisory-level ramp of the combined series was measured again in the two registers separately, BDS 5,0 alone and BDS 6,0 alone, over the same times. Wherever both had data at both ends (43 ramps, 20 of them only found by v8), both registers saw the change with the same sign and at least half its size (median ratios 0.78–1.05). The landings newly flagged by v8 were turbulent: median EDR 0.155 (86 % light or more), against 0.167 for those flagged by both and 0.099 for the unflagged ones.

Table 11. Ramp levels of the stored landings 24 October 20:41 – 30 October 09:00 UTC (1 049 with a v8 result), the advisories relative to the reference flights as in v7.

	none	ADV	WSA
v7: TAS − GS of BDS 5,0	1 016	30	1
v8: BDS 5,0 and BDS 6,0 against the ADS-B GS	1 008	41	0
flagged by both / v8 only / v7 only	17 / 23 / 14 (+2 by the band comparison)		

15.3 Change: v8 in shadow

Why the change. More and less noisy headwind values resolve the short ramps that the ICAO and HKO practice weighs most (the 1-km scaling favours them), and the extra advisories are seen by two independent registers and come with turbulent air. But the advisory rate rises by about a third, and the change touches every flight; one week of the two side by side, live, is the test before v8 alerts.

From 30 October 2026 the detection **v8** (`collector/shear.py`, `analyze_v8`) runs beside v7 for every landing and departure: the headwind from BDS 5,0 TAS and from the BDS 6,0 indicated airspeed converted to TAS, both against the ADS-B ground speed of 0.5 s before the reply, the BDS 6,0 offset of the flight removed, BDS 6,0 replies taking the roll of a BDS 5,0 reply within 4 s; then the same profile, ramps, scaling, levels and band comparison as v7. Its result is stored beside the v7 report (`shear_json "v8"` of a landing, `"trial8"` of a departure; back-filled once for the stored flights with ADS-B ground speed) and shown in the Approach History's hover; it raises no alert. Without ADS-B ground speed a flight gets no v8 result. After about a week the two are compared on the live flights, and v8 replaces v7 if it holds.

16 Results so far

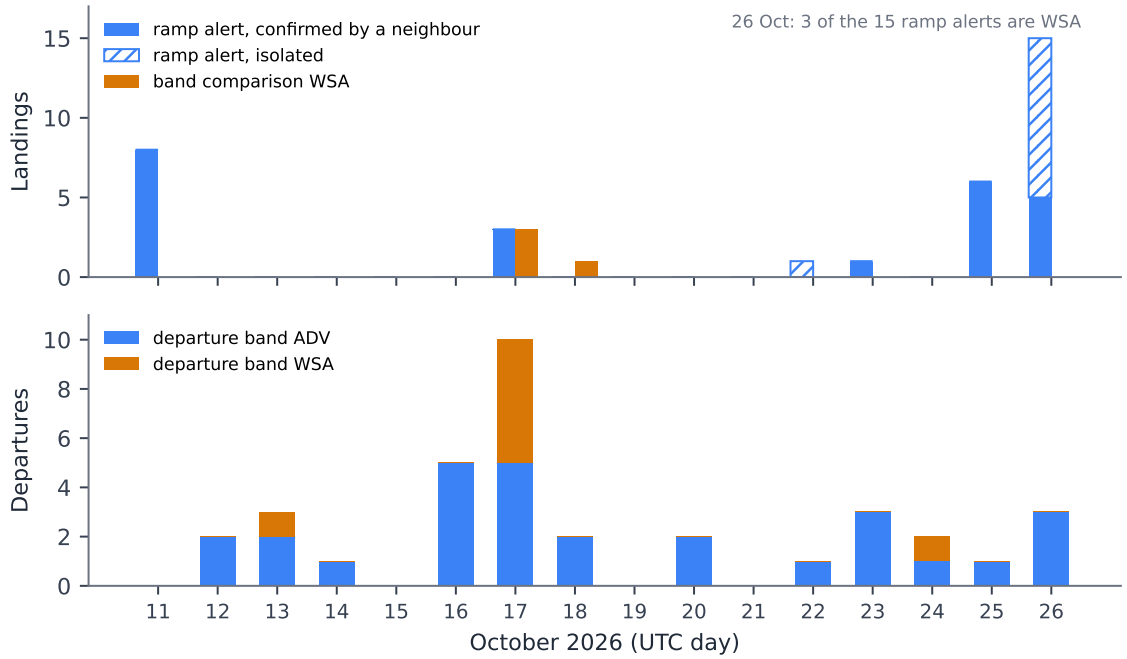


Figure 14. Windshear flags per UTC day with the method v5, 11–26 October 2026 (departure reports start on 11 October). Top: landings, ramp alerts split into confirmed and isolated, and band-comparison WSAs. Bottom: departure band flags.

Table 12. All stored flights with the current method (v7), 9 October – 27 October 2026 18 UTC. With v6 the landing ramps had 41 ADV and 3 WSA.

	Flights	ADV	WSA	MBA
Landings, ramps	3 913	43 (1.10 %)	1 (0.03 %)	0
of which confirmed / isolated		29 / 14 (+1 open)		
Landings, band comparison	3 913	1 (0.03 %)	4 (0.10 %)	—
Departures, band comparison	3 698	21 (0.57 %)	5 (0.14 %)	—

Since 29 October each flight also has its turbulence (EDR; Section 12) and each hour its boundary layer (Section 13). In 9–29 October the windshear levels came with turbulent air (median EDR 0.145–0.16 against 0.065) and were rarer in the stable hours (0.5 % of the landings during a low-level jet, none under a surface inversion, against 1.2–1.3 % otherwise).

The period so far contains no convective weather, and no windshear was reported by METAR in it, so the numbers above describe the method in autumn boundary-layer weather: stable nights, a frontal passage (23–24 October), one sharp wind change (17 October) and two turbulent, windy days (26–27 October).

17 Database checks

Every check is made on a read-only copy of the station database (a "snapshot"), with the logs and the latest soundings. Newest last.

Date	Check and main results
24 Oct 2026	**Restudy of the v1 algorithms** (snapshot 23 Oct; 3 084 landings 9–23 Oct). Noise floor, coverage and false-alarm replay (Tables 2 and 3). Literature review: no published windshear study on passively received Mode S EHS. BDS 4,5 hazard flags found unreliable. ⇒ v1 retired, v2 designed.
24 Oct 2026	**Post-front check** (snapshot 24 Oct 16 UTC). 350 landings: 7 ramp advisories, all gains of 10–11 kt per km on 04L/R on 23 Oct 11:34–14:15 (a persistent shear layer at 1 400–1 800 ft, seen by the other landings at 6–9 kt). Runway changes (04 → 15 → 22) gave no false alarms. The TAS − GS band comparison could not compare 55 % of departures. ⇒ v3 (vector bands, departures).
25 Oct 2026	**Post-deployment check** (night of 24–25 Oct). 98 landings, 109 departures, all levels none; F_h median 0.012, p99 0.026; noise 1.24 kt. ADS-B ground speed collected in the approach series: median 376 messages vs 100 BDS 5,0 replies per landing; BDS 5,0 GS minus ADS-B GS +0.47 ± 0.76 kt.
26 Oct 2026	**Data check** (snapshot 26 Oct 14 UTC). 59 ramp advisories 9–26 Oct (1.6 %), 10 of 39 landings on the stable evening of 25 Oct. ACI rises monotonically with the wind (median 1.1 below 8 kt, 6.1 at 18 kt and gusts). ⇒ v4.
27 Oct 2026	**Data check** (snapshot 27 Oct 05 UTC). The turbulent southerly day: 12 landing ramp alerts in 24 h, one real shear layer (FIN6KB), several isolated eddies. Departures read about +2 kt more headwind than landings over ∼11 000 band comparisons. MEPS check at the aircraft positions and the soundings normal. ⇒ confirmation label.
27 Oct 2026	**Phase-offset analysis** (same snapshot). Acceleration and geography ruled out; the offset is symmetric about MEPS (−1.45 / +1.38 kt). ⇒ v5; replay in Table 6.
27 Oct 2026, evening	**Data check** (snapshot 27 Oct 18 UTC, the first downloaded gzip-compressed from the station). South-westerly flow with a low-level jet (34–40 kt at 0.75–1.1 km), ACI median 3.8. 156 landings since 02 UTC: 11 ADV, no WSA; ramp alerts 6 confirmed, 3 isolated, 1 open; one band ADV (JAL47, a B787 reading 207°/21 kt where its neighbours had 223–228°/26–32 kt — one aircraft). 167 departures: 3 ADV, all ATR 72. Soundings: 12 UTC T bias −0.03 K (RMS 0.44 K), the best launch so far. *Data quality:* on the station the magnetic declination model was missing from 12:54 to 18:25 UTC (fixed 10.5°; exact at EFHK, so no effect on the windshear analysis; up to 2° off far from the airport). ⇒ type and turboprop replay.
27 Oct 2026, evening	**Type-bias and turboprop replay** (same snapshot; every band, 13–27 October, offsets learnt on the first half and tested on the second). Per-type offsets small and repeatable; turboprop departures scatter 1.3 times as much as jet departures. ⇒ v6; replay in Table 7.
28 Oct 2026	**Data check** (snapshot 28 Oct 06 UTC [21]). All reports on v6; turboprop departures now flag 1.2 % (9 of 732), jets 0.7 %. Climb-out series stored for all 61 departures since the deploy (about 2.7 kB each). Night of 27–28 October: south-westerly low-level jet, landings 15, departures 22L/R; 79 landings without a flag, 75 departures with 3 band ADV (Section 9); the first trial ramps flag 4 of 57 climb-outs: two of them the jet itself, one an ATR climb, one a short ramp twice the profile. Sounding 00 UTC T bias −0.6 K (11 aircraft, night); MEPS 7 days normal. ⇒ two questions for the departure ramps.
28 Oct 2026, evening	**Data check and HKO comparison** (snapshot 28 Oct 19 UTC [24]). 18 restarts during the day, all deploys, no errors; no windshear alert since 06 UTC. Replay of the 14-kt rule over 4 155 landings (Table 9): ramp WSAs 3 → 1. Woodfield diagram: 71 long ramps (≥ 3 km, ≥ 14 kt) none of them a WSA. Alert duration: ADV or WSA 2.9 %, WSA 0.34 % of the time (Section 10). ⇒ proposal for v7.

Date	Check and main results
29 Oct 2026	**EDR, boundary layer and departure series** (snapshot 29 Oct 14:41 UTC [30]). EDR method v2 on 4 255 landings (Section 12): ACI 0.66 / 0.67, daily 0.98, METAR wind 0.68, class-independent. Low-level jet 10 % and surface inversion 15 % of the hours (Section 13); windshear levels 0.5 % of the landings during a jet, none under an inversion. Departure EDR: 34 % coverage; the wind vector and the turning pieces do not agree with the straight pieces (0.28 / 0.33). ⇒ climb-out series to 5 000 ft.
30 Oct 2026	**ADS-B ground speed and BDS 6,0** (snapshot 30 Oct 09:12 UTC [31]; 1 062 landings since 24 Oct 20:41). Timing −0.5 s, BDS 5,0 GS vs ADS-B 0.92 kt RMS; BDS 6,0 vs BDS 5,0 headwind bias 0.02, RMS 0.98 kt; 2.3× the values, noise 1.59 → 1.48 kt in the analysis; ramp advisories 30 + 1 WSA → 41, the new ones seen by both registers and on turbulent approaches (Section 15). ⇒ v8 in shadow.

18 Current method and parameters (version v7)

Table 14. Parameters of the detection on 30 October 2026.

Item	Value
Headwind	TAS − GS of each BDS 5,0 reply; drop roll > 5°, repeated registers (< 6 s), > 12 kt from the ±7.5-s median
Profile	distance from integrated GS; 400-m running mean on a 100-m grid
Ramps	L = 400–4 000 m; $\Delta V\,(1\,\mathrm{km}/L)^{1/3}$; coverage ≥ 60 %; ≥ 5$\sigma$; short ramps (≤ 600 m): 3 replies per end, persist 400 m
Zone	below 1 600 ft above the threshold or within 5 NM
Ramp levels	ADV ≥ 10 (10–12 only if ≥ 10 beyond the reference profile), WSA ≥ 15, MBA ≥ 30 (loss) kt per 1 km; a WSA or MBA also needs $\lvert\Delta V\rvert \geq 14$ kt (v7)
Confirmation	same runway direction, ±15 min, same kind, ≥ 7 kt per 1 km, middle within ±300 ft; label only
F-factor	horizontal term, mean over 1 km (reported, not alerted)
Band comparison	200-ft bands 400–1 800 ft; wind vector; median of the flights of the last 45 min (≥ 3); same phase if ≥ 5, else all with the phase bias removed (−1.45 / +1.38 kt); ADV ≥ 10, WSA ≥ 15 kt; a turboprop's departure ADV ≥ 13, WSA ≥ 19.5 kt (limits × 1.3)
Live operation	server-side every 3-s sweep; two analyses to raise an alert; an alert holds 10 min; 24-h log
v8 (shadow, from 30 Oct)	headwind from BDS 5,0 TAS and BDS 6,0 IAS → TAS (pressure altitude, ISA + the flight's own deviation, else of the flights within ±1 h) against the ADS-B GS at $t - 0.5$ s; per-flight BDS 6,0 offset removed; otherwise as above; stored, no alert

19 Limits, open questions and next steps

19.1 Known limits

- **Coverage:** the last 0.6–2.3 NM before the threshold (below 230–730 ft, depending on the runway) is not seen from the receiver site. Microburst outflows are strongest there.
- **The vertical term** of the F-factor ($-w/V$, the downdraft) cannot be observed.
- **After the fact:** every report describes air an aircraft has already flown through; an event can fall between two landings (one every 2–3 minutes in busy hours).
- **Scales below about 500 m** are under the sampling scale of the replies; the reactive on-board warnings in QAR studies act at about this scale [16].

19.2 Next steps

1. **ADS-B ground speed** — *done 30 October* as v8 in shadow (Phase 7). Compare v8 with v7 on the live flights after about a week (the rate of 400–600 m advisories, the confirmation by neighbours, the EDR of the flagged landings, the departures' trial ramps); then switch, or tune the short-ramp rules. Later: the EDR from the denser series (its noise floor N_0 to be derived again).
2. **Convective season 2027:** the real test. Gust fronts and microburst outflows will show whether the WSA and MBA levels and the confirmation rule hold.
3. **Validation** against the stored METAR windshear groups (with the history from 2018) and the pilot reports (Section 14), and go-arounds (detected automatically): hit / miss / false alarm per runway and hour, once there are enough cases.
4. **EDR:** calibrate against the turbulence pilot reports near the airport; check the departures' coverage with the 5 000-ft series after a week; keep the EDR of the flagged landings as a standing measure (confirmed against isolated alerts).

5. **Boundary layer:** read every new flag against the jet / inversion of its hour; compare the WXREP "inversion" reports with the detected inversions; add the sea breeze in spring 2027.
6. **Departure ramps:** the trial runs from 27 October (Section 8); check the stored climb-outs after one to two weeks — the effect of the acceleration on TAS − GS and the noise with few replies — before they may alert. The first night (Section 9) adds two questions: an expectation above the low-level zone (references up to about 3 000 ft, or ramps cut at the top of the zone), and whether a headwind gain on a climb-out should flag at all.
7. **Turboprop limits and type offsets:** review the factor 1.3 when more turboprop departures (and other types than the ATR 72) are stored; decide whether to remove the small per-type offsets (up to 1.4 kt) from the band winds.
8. **Space–time diagram** — *done 27 October* (Section 8); use it on the next fronts and convective days.
9. **Level review** after the first months, and an **EFHK low-level wind climatology** from the stored band profiles.
10. **The 14-kt rule (v7)** — *done 28 October* (Phase 6); review it in the convective season. Keep the alert duration and the Woodfield diagram as standing measures of every later replay.
11. **BDS 4,5**: a validated decoding of the aircraft's own hazard flags would add the state of the on-board reactive system.

References

[1] ICAO. *Annex 3 to the Convention on International Civil Aviation: Meteorological Service for International Air Navigation*. Montréal (wind shear warnings: runway level to 500 m / 1 600 ft).

[2] ICAO (2005). *Manual on Low-Level Wind Shear and Turbulence*, Doc 9817, 1st ed. Montréal.

[3] ICAO. *Technical Provisions for Mode S Services and Extended Squitter*, Doc 9871. Montréal.

[4] Shun, C.M., Chan, P.W. (2008). Applications of an infrared Doppler lidar in detection of wind shear. *J. Atmos. Oceanic Technol.* 25(5), 637–655. doi:10.1175/2007JTECHA1057.1

[5] Chan, P.W., Hon, K.K., Shin, D.K. (2011). Combined use of headwind ramps and gradients based on LIDAR data in the alerting of low-level windshear/turbulence. *Meteorol. Z.* 20(6), 661–670. doi:10.1127/0941-2948/2011/0242

[6] Hon, K.K., Chan, P.W. (2021). Improving lidar windshear detection efficiency by removal of "gentle ramps". *Atmosphere* 12(11), 1539. doi:10.3390/atmos12111539

[7] Lau, S.Y., Choy, B.L. (2005). Wind shear alerting at Hong Kong International Airport. WMO Workshop on Value Added Services for Air Navigation, Toulouse. Hong Kong Observatory reprint 604.

[8] Lewis, M.S., Robinson, P.A., Hinton, D.A., Bowles, R.L. (1994). *The relationship of an integral wind shear hazard to aircraft performance limitations*. NASA TM-109080.

[9] Proctor, F.H., Hinton, D.A., Bowles, R.L. (2000). A windshear hazard index. *9th Conf. on Aviation, Range and Aerospace Meteorology*, Orlando, Amer. Meteor. Soc., 482–487.

[10] Cole, R.E., Todd, R.F. (1996). A comparative performance study of TDWR/LLWAS 3 integration algorithms for wind shear detection. *Workshop on Wind Shear and Wind Shear Alert Systems*, Oklahoma City, Amer. Meteor. Soc.

[11] Haverdings, H., Chan, P.W. (2010). *Quick Access Recorder (QAR) data analysis software for studies of windshear, turbulence and wake vortex*. NLR-TP-2010-435.

[12] de Haan, S. (2011). High-resolution wind and temperature observations from aircraft tracked by Mode-S air traffic control radar. *J. Geophys. Res.* 116, D10111. doi:10.1029/2010JD015264

[13] Müller, M., et al. (2017). AROME-MetCoOp: A Nordic convective-scale operational weather prediction model. *Weather and Forecasting* 32(2), 609–627.

[14] ATC Network (2012). Finavia enhances surveillance capabilities through COMSOFT WAM network. 13 March 2012.

[15] Frequentis (2022). Finland transforms air traffic surveillance with Frequentis' highly resilient Wide Area Multilateration system. Press release, 20 June 2022.

[16] MODE-S Wind project (2026). *Low-level windshear from passively received Mode-S EHS: literature status, a review of the six algorithms and a noise-floor and false-alarm test on 3 084 Helsinki-Vantaa landings*. Internal study, 24 October 2026 (including Hong Kong, China: Deriving and using wind and temperature data from Mode S DAP and ADS-B data, ICAO APAC MET SG/27 IP/10, 2023).

[17] MODE-S Wind project (2026). Post-front check of windshear v2, 24 October 2026. Internal note.

[18] MODE-S Wind project (2026). Data check of 26 October 2026. Internal note.

[19] MODE-S Wind project (2026). Data check of 27 October 2026. Internal note.

[20] MODE-S Wind project (2026). Data check of 27 October 2026, evening, and the type-bias / turboprop replay. Internal notes.

[21] MODE-S Wind project (2026). Data check of 28 October 2026 (snapshot 06 UTC): the first night of the departure trial. Internal note.

[22] Hong Kong, China (2023). *Deriving and using wind and temperature data from Mode S DAP and ADS-B data in Hong Kong, China*. ICAO APAC MET SG/27, Information Paper 10, Bangkok, 4–8 September 2023.

[23] Hong Kong, China (2026). *Hong Kong Observatory's experiences on collection, dissemination and utilisation of aircraft observations on turbulence*. ICAO APAC MET SG/30, Information Paper 05, Bangkok, 20–24 July 2026.

[24] MODE-S Wind project (2026). Data check of 28 October 2026, evening (snapshot 19 UTC), and the replay of the HKO 14-kt rule. Internal note.

[25] Laakso, O. (2026). *MODE-S Wind — Windshear detection v1: the six browser algorithms, in use May – 24 October 2026*. doc/Windshear_Algorithms_v1_Legacy.pdf.

[26] Laakso, O. (2026). *Windshear Detection — User guide*, version 2.8, 29 October 2026. doc/Windshear_User_Guide.pdf.

[27] Kopeć, J.M., Kwiatkowski, K., de Haan, S., Malinowski, S.P. (2016). Retrieving atmospheric turbulence information from regular commercial aircraft using Mode-S and ADS-B. *Atmos. Meas. Tech.* 9, 2253–2265. doi:10.5194/amt-9-2253-2016

[28] Baas, P., Bosveld, F.C., Klein Baltink, H., Holtslag, A.A.M. (2009). A climatology of nocturnal low-level jets at Cabauw. *J. Appl. Meteor. Climatol.* 48(8), 1627–1642. doi:10.1175/2009JAMC1965.1

[29] Finnish Meteorological Institute (2021). *Lentosääpalvelut Suomessa* (Aviation weather services in Finland), 01-2021, §2.1.5–2.1.6 and the WXREP format.

[30] MODE-S Wind project (2026). EDR and low-level jet study, inversion study and departure-EDR tests, 29 October 2026 (snapshot 14:41 UTC). Internal notes.

[31] MODE-S Wind project (2026). ADS-B ground speed and BDS 6,0 airspeed for the windshear headwind: feasibility check, 30 October 2026 (snapshot 09:12 UTC). Internal note.